

Clarity Scheduling System

Business blueprint, discovery guide, and implementation rulebook

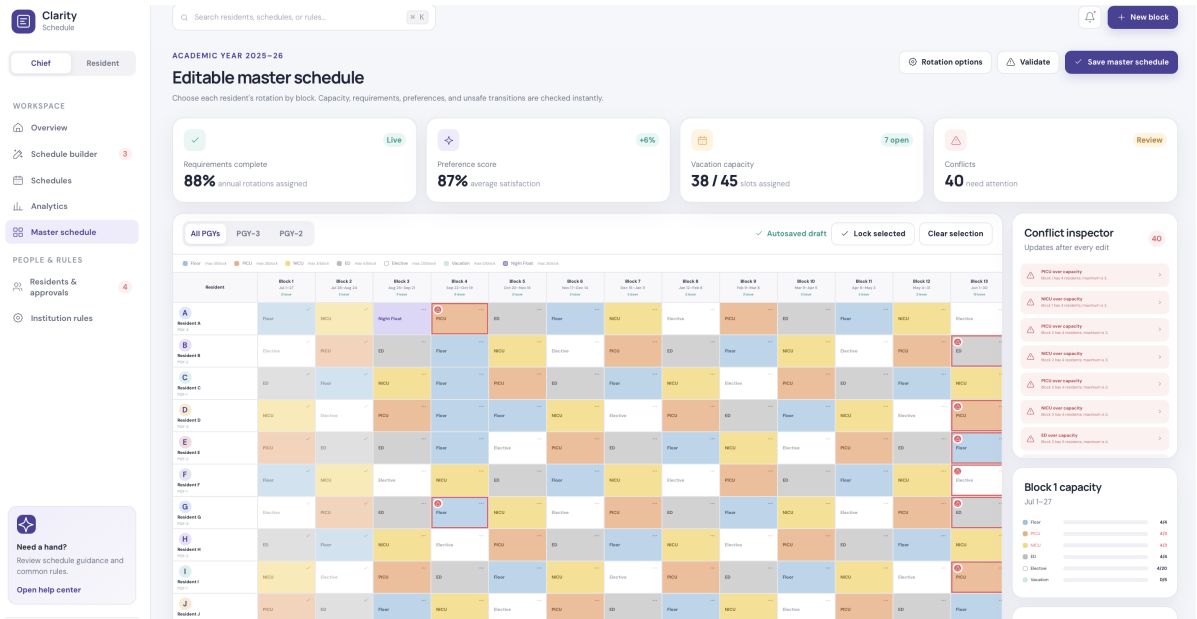
Purpose

A detailed guide for sales, IT, chiefs, coordinators, and hospital leadership to understand what the platform builds, what rules must be collected, and what screens must be linked.

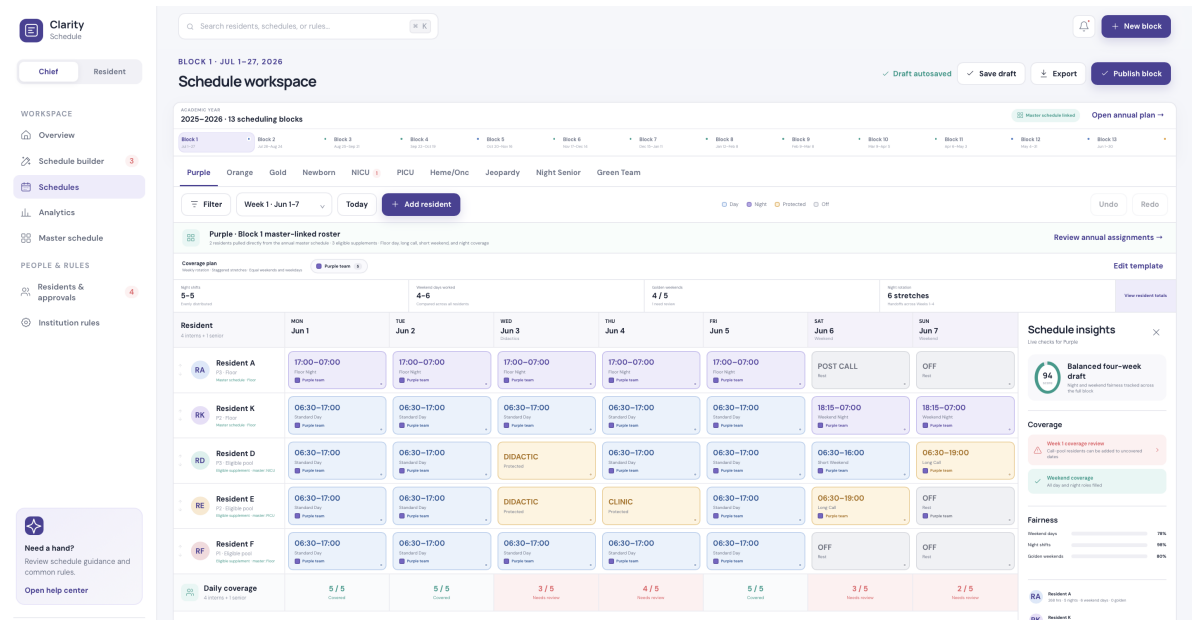
Step	Workflow	What must be linked
1	Annual program setup	Define academic year, block count, dates, PGY classes, tracks, institutions, holidays, rotations, services, roles, minimums.
2	Resident intake	Residents submit fixed facts and annual preferences: master ranking, vacation, holidays, electives, fellowship timing, life events.
3	Chief review and approval	Chiefs review requests, resolve conflicts, approve or decline requests, and mark what can be used by the generator.
4	Master schedule creation	System proposes assignments by PGY and block using rankings plus service minimums; chiefs edit and finalize.
5	Service coverage workbook	Final master becomes service coverage lists by block and service, including outside rotators and supplements.
6	Call pool and holidays	Elective/call-eligible residents flow into call pools; approved holiday requests guide holiday and break coverage.
7	Monthly service schedule builder	For each block and service, the system pulls residents from master/service coverage, protects clinics/didactics/PTO, and drafts editable schedules.
8	Publish and operate	Residents see personal schedules; chiefs monitor stats, switches, PTO, attendance, alerts, and changes.

Visual Examples for Sales and IT Conversations

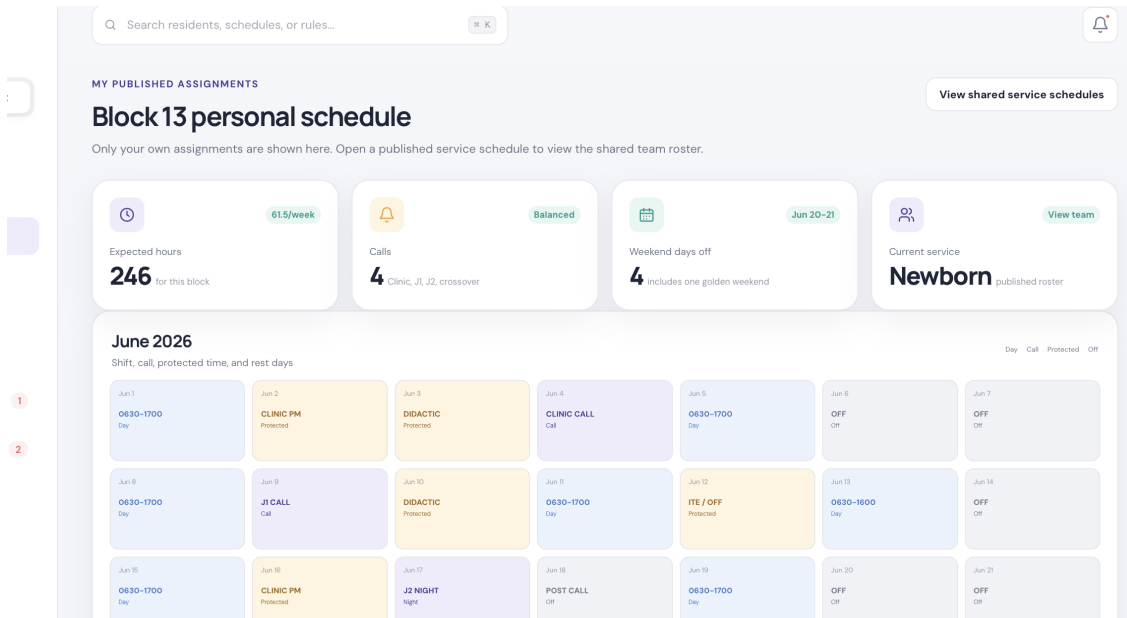
These screenshots are included to make the discussion concrete. They show the design direction: stepwise workflows, chief and resident portals, linked profiles, schedule workspaces, request approvals, and readiness views.



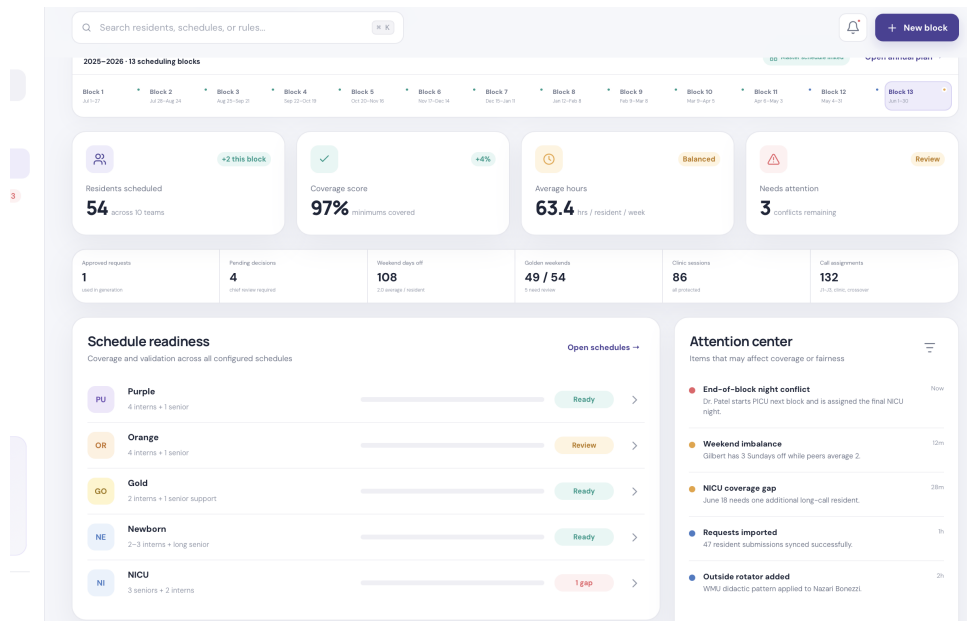
Chief master schedule view: capacity, conflicts, editable annual source data, and profile-linked residents.



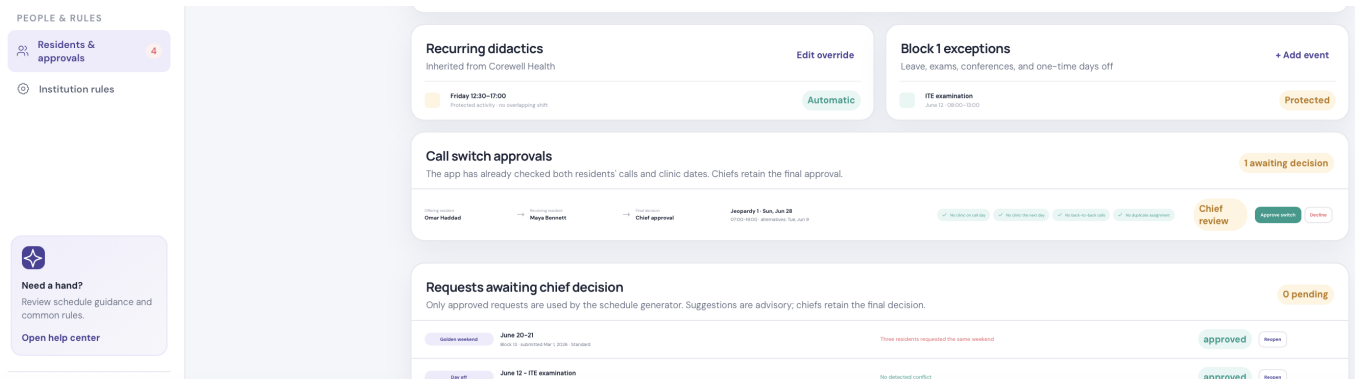
Monthly schedule builder: service tabs, resident rows, shift cards, coverage alerts, and editable drafts.



Resident portal: personal schedule, protected time, calls, master schedule access, requests, and PTO areas.



Chief overview: block readiness, pending risks, coverage, fairness, and attention center.



Approval center: resident requests, call switch checks, and chief decision workflow.

Product Thesis

Layer	What it means
Core scheduling engine	Master schedules, service coverage, call pools, holiday coverage, monthly service schedules.
Workflow tools	Request approval, call switches, PTO/sick leave, attendance, publishing, audit trail.
Decision support	Coverage gaps, over-capacity, fairness, hours, weekends, calls, missed didactics, conflict warnings.

Core Modules and Ownership

Module	Purpose	Feeds / depends on
Resident portal	Resident-facing input and visibility	Feeds resident profile, chief approvals, master builder, monthly scheduler
Chief portal	Command center for building and approving	Feeds every builder and published output
Resident profile	Single source of resident-specific facts	Used by master, service builder, call switch checks, stats
Institution rules	Reusable patterns for home and outside institutions	Auto-protects time when residents are scheduled
Master schedule builder	Annual source of truth	Outputs rotation assignment per resident per block
Service coverage workbook	Bridge from master to monthly schedules	Outputs who covers each service each block
Schedule builder	Monthly schedule generator/editor	Outputs editable and publishable service schedules
Call pool workbook	Call-eligible resident pools	Outputs clinic call, jeopardy, cross-cover, no-call lists
Holiday workbook	Holiday coverage planning	Outputs holiday staffing and fairness stats
Attendance	Didactic check-in and tracking	Updates chief and resident attendance stats
Requests and approvals	Workflow engine for decisions	Approved items become scheduling constraints
Analytics and alerts	Decision support	Shows fairness, gaps, risks, and action items

Critical Linkage Map

Data / action	Starts in	Must update	Business rule
Annual master ranking	Resident portal	Chief master builder	Preferences shown by submission order, template rank, fellowship/life-event flags; chiefs can approve or override.
Vacation and holiday requests	Resident portal	Master builder, holiday workbook, resident profile	Only approved requests become protected constraints; pending requests remain warnings.
Fellowship/maternity/life events	Resident portal/profile	Master builder and chief review	Used as decision support, not automatic final authority.
Clinic pattern	Institution profile or resident profile	Monthly schedule builder, call switch checker	Protects clinic time and prevents calls on clinic day or before clinic when rule applies.
Didactics rules	Institution profile or resident profile	Monthly builder and attendance	Defines required attendance, virtual/in-person mode, and protected time.
Final master assignment	Chief master builder	Service coverage, schedule builder, resident view	Primary source for who belongs on each service in each block.
Service coverage roster	Service coverage workbook	Monthly schedule builder	Pulls names into service tabs like PICU, NICU, Purple, Orange, Gold, etc.
Approved PTO/sick/leave	Resident/Chief request workflow	Resident profile, schedule builder, analytics	Blocks time, updates leave history, identifies service impact.
Call switch offer	Resident portal	Other residents, chief approval, schedule builder	Eligibility checked automatically; final switch requires chief approval.
Attendance checkmark	Attendance tab	Resident profile and chief analytics	Updates attended/missed counts by week, block, year, and requirement type.

Discovery Checklist for Any Department

Category	Questions to ask
Program structure	How many blocks per year? What are block start/end dates? Are there 13 four-week blocks, 26 two-week blocks, monthly blocks, or mixed blocks? Do holiday breaks split blocks?
Training classes	Which PGY classes exist? PGY-1, PGY-2, PGY-3, fellows, Med-Peds, off-cycle residents, outside rotators? How many residents per class?
Tracks	Do tracks change required rotations? Examples: primary care, critical care, hospitalist, subspecialty, research, chief, Med-Peds.
Core rotations	Which rotations are mandatory per PGY? How many times must each resident complete each? Which PGY levels are eligible?
Electives	Which rotations are elective? Which are vacation-eligible or call-eligible? Which are advanced electives? Which PGY levels can request them?
Master templates	Does the program use lettered templates A-T or individualized assignment? How many templates are available? Do residents rank templates?
Capacity rules	For each block, what is the minimum and maximum number of residents needed per service? Are these fixed or census-dependent?
Requests priority	What is the priority logic: first come first serve, fellowship timing, maternity/major life event, seniority, historical fairness, chief discretion?
Protected time	What clinic, didactic, conference, ITE, retreat, exam, and administrative days must be protected? Are they institution-specific?
Night rules	Who can do nights? Max consecutive nights? Post-call rules? No inpatient after night block? Weekend night rules? Split night blocks?
Call rules	What types exist: clinic call, J1/J2/J3 jeopardy, crossover, procedure call, long call, short call, 24-hour call? Who is eligible?
Weekend fairness	How many Saturdays, Sundays, worked weekends, and golden weekends should each resident receive per block and year?
Holidays	Which holidays matter? How do residents rank holiday requests? What minimum staffing is required per holiday and service?
Attendance	Which conferences require check-in? Are sessions required for all residents or only certain rotations? Is virtual attendance allowed?
PTO and leave	How are PTO, sick, maternity, conference, interview, and LOA tracked? What approval steps are needed? What reports does leadership need?
Publishing	What is public to all residents? What is private? Can residents see only their master schedule but all published service schedules?
Audit and compliance	What changes require notes? Who can override warnings? What reports are needed for GME, leadership, or ACGME-style monitoring?

Master Schedule Rules Checklist

Rule area	Detailed rule / question
PGY-specific curriculum	Core rotations differ by year. Example concept: PGY-1 has many core floor/outpatient blocks; PGY-2 may add ICU/night float; PGY-3 may have advanced electives and leadership/board review.
Core vs elective classification	Every rotation must be tagged as core, elective, vacation-eligible, call-eligible, no-call, ICU, floor, ED, outpatient, or custom category.
Template ranking	Residents rank master schedule templates. The system must enforce unique ranks and show chiefs conflicts when multiple residents prefer the same template.
Preference priority	Resident preferences guide the draft, but chiefs need final control. Suggested ranking signals: submission timestamp, fellowship application, major life event, seniority, and fairness.
Service minimums	Each block must meet service minimums such as Purple/Orange, NICU, PICU, ED, Cardiology, Newborn, Heme/Onc, floor senior, gold senior, etc.
Vacation capacity	Vacation-eligible blocks must have maximum allowed vacation slots; the system should flag blocks with too many vacation requests.
Unsafe transitions	If a resident ends a block on nights, ICU nights, NICU nights, floor nights, or night float, the next block should avoid inpatient start if the program requires post-call recovery.
Outside rotator planning	If a service is below minimum, the system should flag need for outside rotators or master schedule adjustment.
Editable final grid	Chiefs must be able to drag/drop, change rotations, lock cells, validate, and save the finalized master as the source of truth.

Annual Resident Intake Forms

Form section	Fields to collect	Where it links
Fixed resident facts	Name, email, PGY, institution, track, fellowship interest, outside rotator status	Resident profile, eligibility checks, institution rules
Master schedule ranking	Rank templates/preferences; enforce unique ranks; timestamp submission	Chief master builder
Vacation/holiday/elective requests	Preferred blocks, holiday rank, top electives, comments	Master builder, holiday workbook, PTO workflow
Life events and special needs	Maternity/paternity, wedding, visa/travel, interviews, board review	Chief review and protected time

Master Builder Design Requirements

Step	Screen goal	Required functions
Choose PGY	Select class/curriculum	Blocks, residents/templates, rotations, min/max, eligible blocks, drag/add/delete rotations
Resident rankings	Review preferences	Sort by submission, alphabet, template; open resident profile
Block coverage	See counts before final grid	Clickable block/service counts, names, min/max, gaps
Finalize master	Editable annual grid	Drag/drop, lock, validate, save source, export Excel-like view
Import existing master	Skip builder	Upload Excel/PDF, parse, map, review, approve import

Monthly Service Schedule Builder Rules

Rule area	Detailed rule / question
Roster pull	When chiefs choose Block X and Service Y, residents assigned to that service in the finalized master should auto-load into the schedule builder.
Supplement pull	Call pool or outside rotator residents can be added as supplements with visible source labels.
Protected time pull	Clinics, didactics, exams, retreats, PTO, sick leave, maternity leave, conferences, and approved special requests should appear before shifts are assigned.
Service-specific rules	Each service has its own staffing minimums, roles, units, shift types, and PGY eligibility. PICU may have Unit 1/Unit 2; NICU may have procedure call; floors may have colored teams.
Shift templates	Day, night, long call, short call, procedure call, post-call, protected, off, crossover, and custom shifts should be configurable by service.
Fairness engine	Track nights, weekends, Saturdays, Sundays, golden weekends, calls, clinic calls, hours, and days off across the block and year.
Chief editability	The system drafts; chiefs edit. Drag/drop, copy, custom time, resident swap, row reorder, undo/redo, and notes are required.
Publish behavior	Published service schedules become visible to residents; personal schedules combine service assignments, calls, clinics, didactics, PTO, and attendance obligations.

Night Shift, Call, and Clinic Protection Rules

Rule family	Specific questions for discovery
Night eligibility	Which PGY levels/services can do nights? Outside rotators? Interns?
Night stretch	Maximum consecutive nights, post-call rules, unsafe next block rules.
Clinic protection	Can calls occur on clinic day or the day before? Does AM/PM matter?
Call categories	Clinic call, J1/J2/J3, crossover, procedure call, long/short call, 24-hour call.
Fairness counters	Calls, weekends, nights, golden weekends, hours, days off.

Call Switch Workflow Rules

Stage	Rules and system behavior
Offer	Resident selects an assigned call and offers it for switch, optionally listing acceptable alternative dates.
Marketplace	Other residents see available calls with eligibility indicators before accepting.
Automatic checks	No clinic on call day, no clinic next day if post-call rule applies, no back-to-back calls, no duplicate assignment, PGY/service eligibility, duty hour risk.
Acceptance	Receiving resident requests to take or swap. The system shows why they are eligible or not eligible.
Chief approval	Chiefs see both residents, call details, clinic/didactic conflicts, back-to-back status, and final recommendation; chiefs approve or decline.
Propagation	Approved switch updates the service schedule, resident personal schedules, call totals, profile histories, and audit log.

Attendance and Didactics

Feature	Details
Attendance tab	Create lecture session by block/date/time/type/requirement/mode.
Resident check-in	Tablet-friendly list of residents with checkmarks.
Profile linkage	Updates resident/chief profile and analytics.
Statistics	Attended/missed by week, block, year, rotation, institution, excused/unexcused.

PTO, Sick Leave, LOA, Conferences, Exams

Leave type	What to collect and track
PTO/vacation	Dates/blocks, priority, approval, service impact, vacation capacity, yearly total.
Sick leave	Date, service, clinic impact, inpatient/elective status, coverage needed, running total.
Maternity/LOA	Date range, blocks affected, privacy rules, coverage impact.
Conference/exam/ITE	Protected day/partial day, travel buffer, coverage need, resident profile update.

Resident and Chief Portal Requirements

Portal	Required areas
Resident	My overview, my schedule, private master, published schedules, requests, call switches, PTO/sick leave, attendance.
Chief	Overview, master schedule, service coverage, monthly builder, residents/profiles, rules/institutions, attendance, analytics.

Statistics and Alerts Catalog

Statistic / alert	Where shown	Why it matters
Residents per service per block	Master, service coverage, overview	Prevents short or overstaffed services.
Outside rotator burden	Service coverage	Prevents overcalling one block.
Weekend/golden weekend fairness	Schedule builder	Keeps fairness visible.
Night/call totals	Profile and analytics	Avoids uneven workload.
Clinic conflicts	Call switch/scheduler	Prevents unsafe assignments.
Attendance missed/attended	Attendance/profile	Supports education tracking.

Design and User Experience Standards

Design requirement	Implementation meaning
Dual experience	Separate Chief portal and Resident portal with different permissions and visibility.
Stepwise builders	Master builder and monthly service builder should walk users through setup, people, protected time, requests, rules, generate, edit, publish.
Digital and Excel-like views	Every major schedule should support modern card/table view plus Excel-like grid view for transition comfort.
Upload-first option	Programs with existing Excel/PDF masters can upload files; the system parses names, PGY, blocks, rotations, service coverage, call pools, holidays, and maps them to profiles.
Clickable details	Counts should be clickable. Clicking a service, block, holiday, call pool, or resident should show names, sources, conflicts, and links to profiles.
Not crowded	Use progressive disclosure: show counts first, then open details on click. Keep side panels for insights instead of forcing all data into one grid.
Visual alerts	Use clear badges: Ready, Review, Gap, Over, Pending, Approved, Declined, Private, Published.
Autosave and audit	Draft autosave, save draft, publish, export, undo/redo, and change notes should be obvious.
Accessibility	Colors should help but not be the only signal. Use labels and status text in addition to color.

Import and Migration Requirements

Upload type	What the system should extract	Human review needed
Existing master Excel	PGY tabs, names, blocks, rotations, colors, notes, vacation columns	Unmatched names, ambiguous rotations, split blocks
Service/call/holiday Excel	Service rows, block columns, names, pools, holiday staffing	Mappings, red/struck names, partial blocks
PDF final schedules	Dates, services, residents, work hours, calls, nights, protected time	PDF parsing needs review

MVP Phasing Recommendation

Phase	Build	Definition of done
Phase 1	Data foundation	Can represent real residents and master assignments.
Phase 2	Master + service coverage	Chiefs validate annual staffing.
Phase 3	Monthly scheduler	One block can be generated, edited, published.
Phase 4	Requests and switches	Approved decisions update schedules/profiles.
Phase 5	Attendance and analytics	Leadership can track education/workload/fairness.

Department Configuration Worksheet

Question	Answer field for discovery notes
Program name and specialty	
Number of blocks and date model	
PGY classes / tracks	
Core rotations by PGY	
Elective/vacation/call-eligible rotations	
Services and locations	
Minimum/maximum staffing	
Shift templates and call types	
Clinic and didactic protection rules	
Night/post-call rules	
Request priority rules	
Holiday and break rules	
Attendance requirements	
Reports/statistics needed	
Preferred design	Digital, Excel-like, or both